STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	3

Work Done Summary

In this case study, we analyzed Aadhaar enrollment connectivity using graph traversal and Depth First Search (DFS).

The work focused on:

1. Representing enrollment centres as graph vertices.
2. Representing shared enrollments as graph edges.
3. Applying DFS to identify connected components.
4. Tracing DFS traversal across all centres.
5. Counting disconnected components in the graph.
6. Calculating component sizes.
7. Identifying isolated centres with no edges.
8. Understanding operational implications of disconnected populations.
9. Studying why larger connected networks improve interoperability and fraud detection.

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Implementation Details

1. Modeled Aadhaar enrolment centres as graph nodes.
2. Represented shared-person relationships as undirected graph edges.

3. Constructed the graph using adjacency lists.
4. Initialized all vertices as unvisited.
5. Applied DFS traversal starting from each unvisited node.
6. Marked visited nodes during traversal.
7. Grouped all reachable nodes into the same connected component.
8. Processed the given graph:

```
c1-c2
c2-c3
c3-c1
c4-c5
c6-c7
c7-c8
```

9. Identified isolated node:

```
c9
```

10. Computed connected components:

- Component 1 → {c1, c2, c3}
- Component 2 → {c4, c5}
- Component 3 → {c6, c7, c8}
- Component 4 → {c9}

11. Counted total connected components:

```
4 Components
```

12. Calculated component sizes:

- Size 3
- Size 2
- Size 3
- Size 1

13. Explained operational impact of disconnected regions.

RESULTS:SCREENSHOTS/Output:

```
Starting DFS Traversal...

Component 1:
c1 → c2 → c3
Size = 3

Component 2:
c4 → c5
Size = 2

Component 3:
c6 → c7 → c8
Size = 3

Component 4:
c9
Size = 1

Total Connected Components = 4

DFS Traversal Completed Successfully.
```

Conclusion

In this case study, DFS was used because it efficiently identifies connected components in undirected graphs.

The Aadhaar enrolment network was divided into four disconnected components, indicating that several groups of enrolment centres do not share enrolment relationships.

Isolated or smaller components may represent geographically separated or operationally disconnected regions.

Larger connected components are desirable because they improve interoperability, data consistency, and fraud detection across centres. DFS-based connectivity analysis helps large-scale government systems identify disconnected regions and improve network integration efficiently.

Github Link: https://github.com/ampssubrahmanyasreepad-svg/DSA-CaseStudies-correct

Student Signature	Faculty Signature
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